

Supplementary Information

Document caption: Detailed methods, supplementary analyses, and results including p-values.

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1 Current climate conditions

Data is available 2007 – 2022. 2008 is the first whole year of temperature data. Data available from <https://data.g-e-m.dk> (GEM 2026f, 2026d).

1-1 Temperature**Monthly mean**

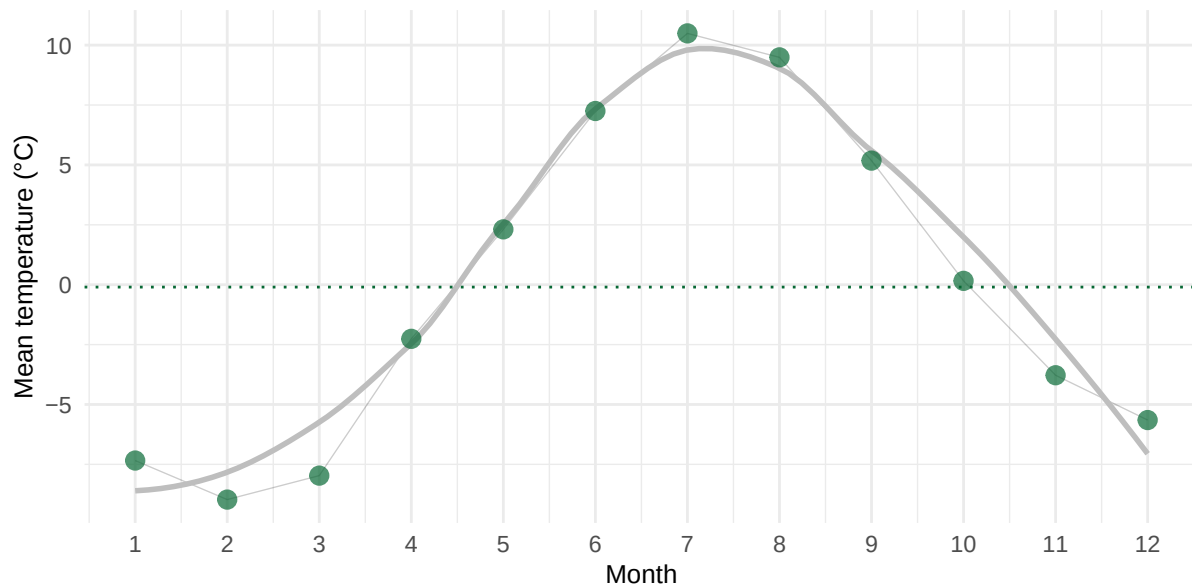


Figure 1: Mean temperature pr month. Mean air temperature (°C) pr month, based on data from 2007 to 2022. The dashed line represent yearly mean of -0.1°C.

Yearly mean

```
model_temp_annual_mean <- lm(mean ~ year, data = data_temp_annual_mean)
summary(model_temp_annual_mean)
```

Call:

```
lm(formula = mean ~ year, data = data_temp_annual_mean)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.4656	-0.6153	-0.1082	0.6452	3.3959

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.679175	170.533389	0.045	0.965
year	-0.003821	0.084632	-0.045	0.965

Residual standard error: 1.416 on 13 degrees of freedom

Multiple R-squared: 0.0001568, Adjusted R-squared: -0.07675

F-statistic: 0.002038 on 1 and 13 DF, p-value: 0.9647

```
data_temp_annual_mean$pred <- predict(model_temp_annual_mean)
```

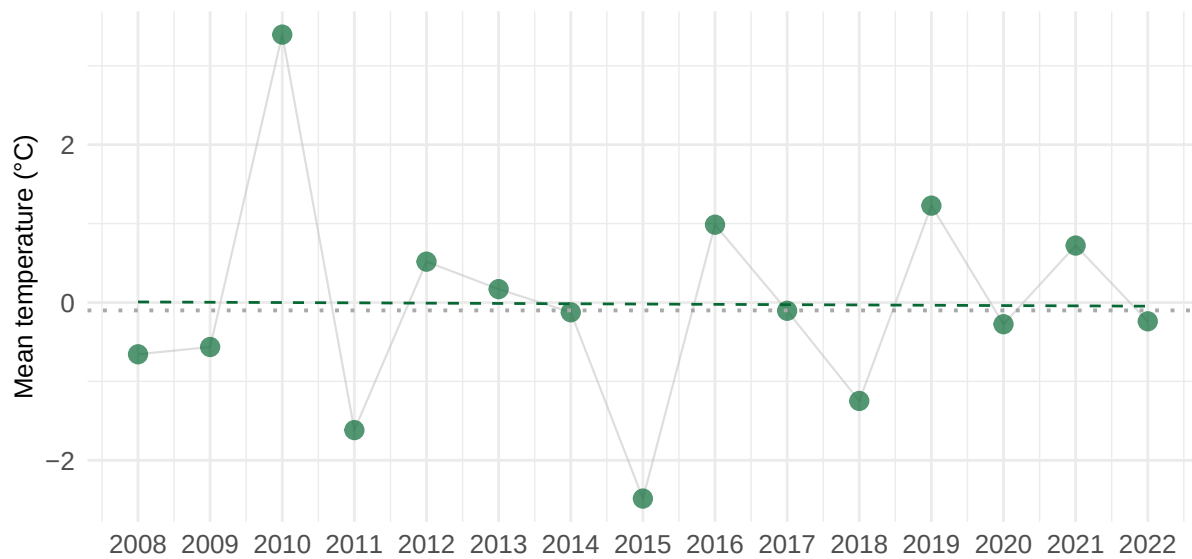


Figure 2: Mean annual air temperature pr year (°C) from 2008 to 2022 in Kangerluassunguaq. Data from 2007 is excluded because data is only from October, November and December. Gray dotted line indicate over all mean of -0.1003°C. Dashed line is model prediction.

Five year intervals

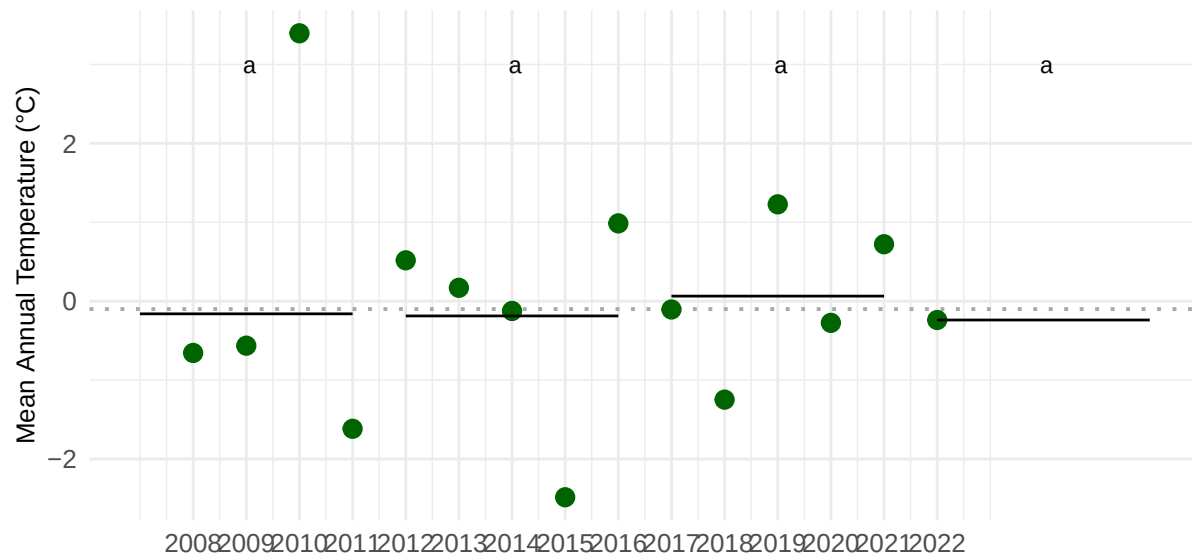


Figure 3: Five year intervals in temperature means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

1-2 Precipitation

Data is available from GEM. (2020). *ClimateBasis Nuuk—Precipitation—Precipitation accumulated (mm)* (Version 1.0) [Dataset]. Greenland Ecosystem Monitoring. <https://doi.org/10.17897/SXJ8-WA79>. Downloaded on XXXX.

The mean annual precipitation is 886.15 mm.

The months with the most precipitation are September (138.64 mm), October (99.38 mm), August (98.28 mm), which account for 336.3 mm or 38 % of the yearly precipitation.

The months with the least precipitation are May (51.51 mm), February (37.62 mm), March (36.44 mm).

Monthly mean

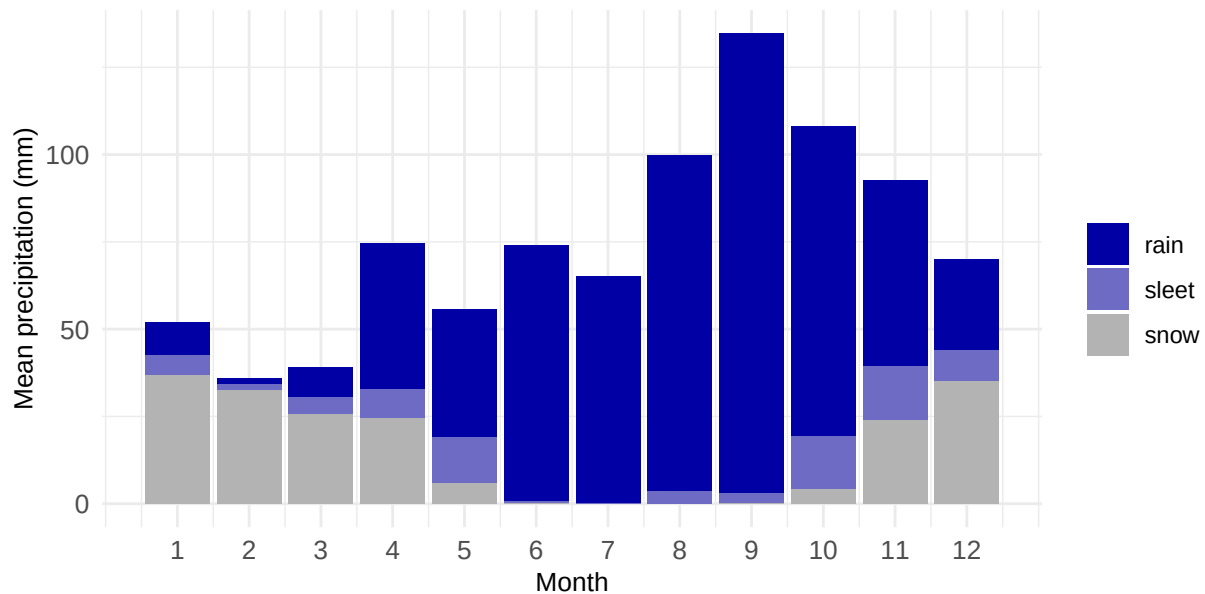


Figure 4: Mean precipitation type pr month. Mean monthly distribution of types of precipitation (mm). Based on data from 2008 to 2022. Sleet is defined as precipitation that fell when temperatures were between -1°C and 1°C .

Yearly mean

Call:

```
lm(formula = yearly_precip ~ year, data = data_precip_mean_annual)
```

Residuals:

Min	1Q	Median	3Q	Max
-405.29	-221.91	54.49	205.78	307.56

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-9763.076	24027.287	-0.406	0.690
year	5.282	11.918	0.443	0.664

Residual standard error: 240.7 on 15 degrees of freedom

Multiple R-squared: 0.01293, Adjusted R-squared: -0.05288

F-statistic: 0.1964 on 1 and 15 DF, p-value: 0.6639

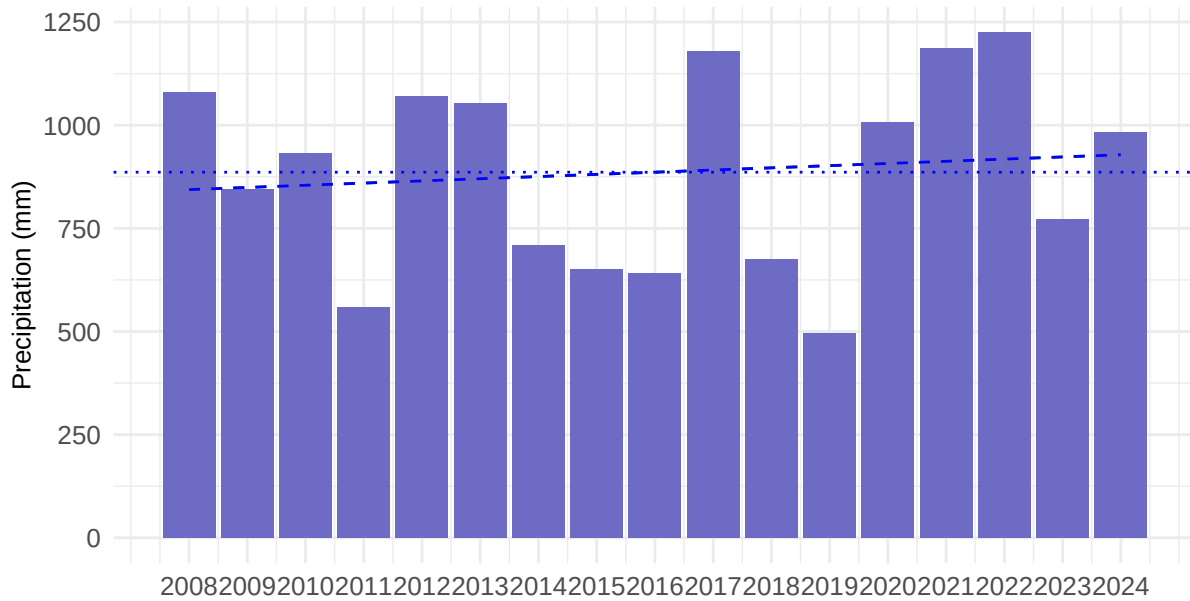


Figure 5: Mean yearly precipitation (mm) from 2008 to 2024. Data from 2007 is excluded because it was only from May - December. Blue dashed line indicate over all mean of 886.15 mm. Dashed line represent model prediction.

Five year interval precipitation

date	time	pre_mm
Min. :2008-01-01	Min. :00:00:00.000000	Min. : 0.0000
1st Qu.:2012-07-23	1st Qu.:06:00:00.000000	1st Qu.: 0.0000
Median :2016-09-30	Median :12:00:00.000000	Median : 0.0000
Mean :2016-08-30	Mean :11:30:21.122933	Mean : 0.1058
3rd Qu.:2020-12-09	3rd Qu.:18:00:00.000000	3rd Qu.: 0.0000
Max. :2024-12-31	Max. :23:00:00.000000	Max. :19.0000

quality_flag	year	month	doy
Length:142395	Min. :2008	Min. : 1.000	Min. : 1.0
Class :character	1st Qu.:2012	1st Qu.: 4.000	1st Qu.: 99.0
Mode :character	Median :2016	Median : 7.000	Median :187.0
	Mean :2016	Mean : 6.625	Mean :186.3
	3rd Qu.:2020	3rd Qu.:10.000	3rd Qu.:276.0
	Max. :2024	Max. :12.000	Max. :366.0

month_string	anchor	seg_start	seg_end
Length:142395	Min. :2007	Min. :2007	Min. :2011
Class :character	1st Qu.:2007	1st Qu.:2012	1st Qu.:2016
Mode :character	Median :2007	Median :2012	Median :2016

	Mean	:2007	Mean	:2014	Mean	:2018
	3rd Qu.	:2007	3rd Qu.	:2017	3rd Qu.	:2021
	Max.	:2007	Max.	:2022	Max.	:2026
segment	seg_mean_precip					
Length:142395	Min.	:2980				
Class :character	1st Qu.	:3413				
Mode :character	Median	:4128				
	Mean	:3877				
	3rd Qu.	:4544				
	Max.	:4544				

1-3 Radiation

Data available from <https://data.g-e-m.dk> (GEM 2026e)

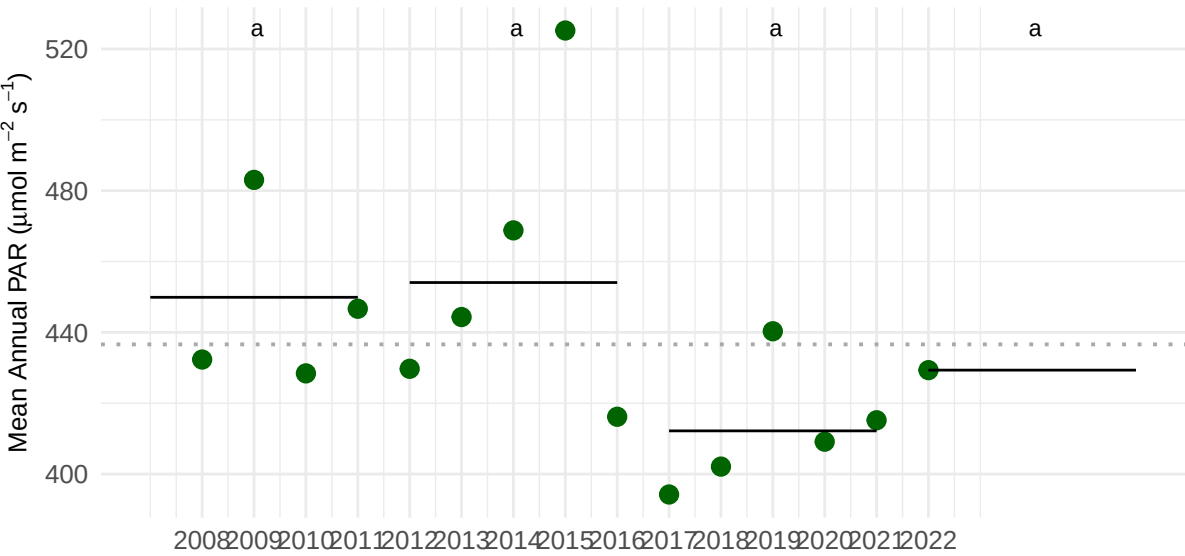


Figure 6: Five year intervals in PAR means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

2 Historic climate data

2-1 Temperature

Data available from Cappelen (2021).
Station 4250 is Nuuk, and element 101 is precipitation.

```
model_annual_temp_historic <- lm(annual ~ year, data = nuuk_21_04_dmi_temp)
summary(model_annual_temp_historic)
```

Call:

```
lm(formula = annual ~ year, data = nuuk_21_04_dmi_temp)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.3697	-0.6837	0.0834	0.7403	3.3848

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-19.764947	2.851721	-6.931	8.06e-11 ***
year	0.009443	0.001478	6.389	1.51e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.163 on 172 degrees of freedom

Multiple R-squared: 0.1918, Adjusted R-squared: 0.1871

F-statistic: 40.81 on 1 and 172 DF, p-value: 1.512e-09

```
nuuk_21_04_dmi_temp$pred <- predict(model_annual_temp_historic)
```

```
preds <- predict(model_annual_temp_historic, se.fit = TRUE)
```

2. Create a new dataframe for the ribbon

We calculate the Upper and Lower bounds (Fit $\pm$ 1.96 * SE for 95% CI)

```
nuuk_21_04_dmi_temp$fit <- preds$fit
```

```
nuuk_21_04_dmi_temp$se <- preds$se.fit
```

```
nuuk_21_04_dmi_temp$lower <- preds$fit - 1.96 * preds$se.fit
```

```
nuuk_21_04_dmi_temp$upper <- preds$fit + 1.96 * preds$se.fit
```

```
ggplot(data = nuuk_21_04_dmi_temp, aes(x = year, y = annual)) +
  geom_line(
    aes(y = pred),
    color = "#076834",
    linetype = "solid",
  ) +
  geom_ribbon(aes(ymin = lower, ymax = upper),
    fill = "#076834",
```

```

      alpha = 0.2) +
geom_point(size = 3, alpha = 0.7, color = "#076834") +
#geom_smooth(method = "lm")+
scale_x_continuous(
  breaks = seq(min(gr_monthly_all_1784_2020$year),
               max(2025), by = 15))+
labs(x = "", y = "Temperature (°C)") +
theme_minimal()+
  theme( axis.title.x = element_text(size = 10),
axis.title.y = element_text(size = 10),
axis.text.x = element_text(size = 10),
axis.text.y = element_text(size = 10))

```

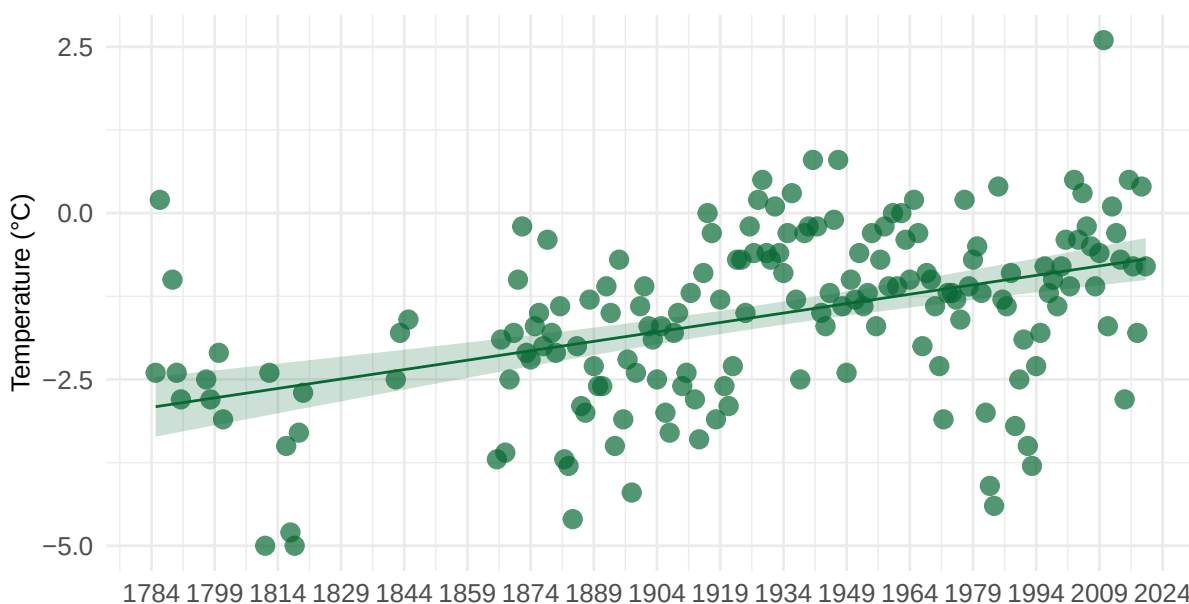


Figure 7: Historic yearly mean air temperature (°C) in Nuuk, from 1785 to 2020. Yearly mean of -1.6°C. Yearly mean ranges from -5°C to 2.6°C. Modelled mean temperature has increase from -2.92°C in 1784 to -0.65°C in 2024 (increase of 2.27°C or 0.095°C decade¹.)

3 Statistical analysis

3-1 Diversity

Data available from <https://data.g-e-m.dk> (GEM 2026a).

Importing data and calculating indexes

```
#reading the raw datafile
data_raw <- readRDS("~/Library/CloudStorage/OneDrive-Aarhusuniversitet/MappingPlants/01 Vegetation")

data_filtered <- data_raw |>
  group_by(year, subsection) |>
  mutate(no_plots = n_distinct(plot_id)) |>
  ungroup() |>
  filter(taxon_code != "rock", no_plots > 1, veg_type != "saltmarsh") |>
  dplyr::select(year, vt_section, subsection, plot_id, no_plots, veg_type, taxon_code, presence)

data_pivotwide <- data_filtered |>
  tidyr::pivot_wider(
    id_cols = c(year, vt_section, subsection, plot_id, no_plots, veg_type),
    names_from = taxon_code,
    values_from = presence,
    values_fill = 0,
    values_fn = max
  )

data_pivotlong <- data_pivotwide |>
  pivot_longer(cols = carbig:viospe, names_to = "taxon_code", values_to = "presence")

#calculation count and frequency of occurrence
subsection_species <- data_pivotlong |>
  group_by(year, subsection, veg_type, no_plots, taxon_code) |>
  summarise(count = sum(presence), foo = count/no_plots) |>
  distinct()

subsection_species_wide <- subsection_species |>
  dplyr::select(-count, -no_plots) |>
  pivot_wider(names_from = taxon_code, values_from = foo, values_fill = 0)

# calculating richness, evenness and diversity
subsection_indices <- subsection_species |>
  dplyr::group_by(year, subsection, veg_type) |>
  dplyr::summarise(
    # restrict to taxa that are present
    H = vegan::diversity(count[count > 0], index = "shannon"),
    S = dplyr::n_distinct(taxon_code[count > 0]),
    shannon = H,
```

```

    evenness = ifelse(S > 1, H / log(S), NA_real_),
    richness = S,
    .groups = "drop"
  )

years <- c("2007", "2012", "2017", "2022")

```

Data has been filtered to remove sections with only one remaining plot and subsections in saltmarsh, due to too few replicates in this vegetation type. Final calculates included 134 subsections (original dataset includes 140).

Richness

```

model_richness_sub_f <- lmer(
  richness ~ factor(year) + (1 | subsection),
  data = subsection_indices
)

summary(model_richness_sub_f)

```

Linear mixed model fit by REML ['lmerMod']
 Formula: richness ~ factor(year) + (1 | subsection)
 Data: subsection_indices

REML criterion at convergence: 2159.7

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.0211	-0.4767	-0.0042	0.4437	5.2716

Random effects:

Groups	Name	Variance	Std.Dev.
subsection	(Intercept)	9.173	3.029
Residual		2.219	1.490

Number of obs: 492, groups: subsection, 134

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	10.022586	0.292140	34.307
factor(year)2012	0.557353	0.185243	3.009

```
factor(year)2017 -0.753326  0.191972 -3.924
factor(year)2022  0.004956  0.195410  0.025
```

Correlation of Fixed Effects:

```
(Intr) f()2012 f()2017
fctr(y)2012 -0.312
fctr(y)2017 -0.302  0.484
fctr(y)2022 -0.296  0.476  0.480
```

```
richness_emmeans_f <- emmeans(model_richness_sub_f, ~ year)

richness_pairs_f <- emmeans(model_richness_sub_f, ~ year) |>
  pairs()

richness_emmeans_f
```

year	emmean	SE	df	lower.CL	upper.CL
2007	10.02	0.292	179	9.45	10.60
2012	10.58	0.293	181	10.00	11.16
2017	9.27	0.297	190	8.68	9.86
2022	10.03	0.299	195	9.44	10.62

Degrees-of-freedom method: satterthwaite
Confidence level used: 0.95

```
richness_pairs_f
```

contrast	estimate	SE	df	t.ratio	p.value
year2007 - year2012	-0.55735	0.185	357	-3.009	0.0148
year2007 - year2017	0.75333	0.192	359	3.924	0.0006
year2007 - year2022	-0.00496	0.195	359	-0.025	1.0000
year2012 - year2017	1.31068	0.192	358	6.838	<0.0001
year2012 - year2022	0.55240	0.195	358	2.831	0.0251
year2017 - year2022	-0.75828	0.198	356	-3.838	0.0008

Degrees-of-freedom method: satterthwaite
P value adjustment: tukey method for comparing a family of 4 estimates

```
richness_pairs_df <- as.data.frame(richness_pairs_f)
```

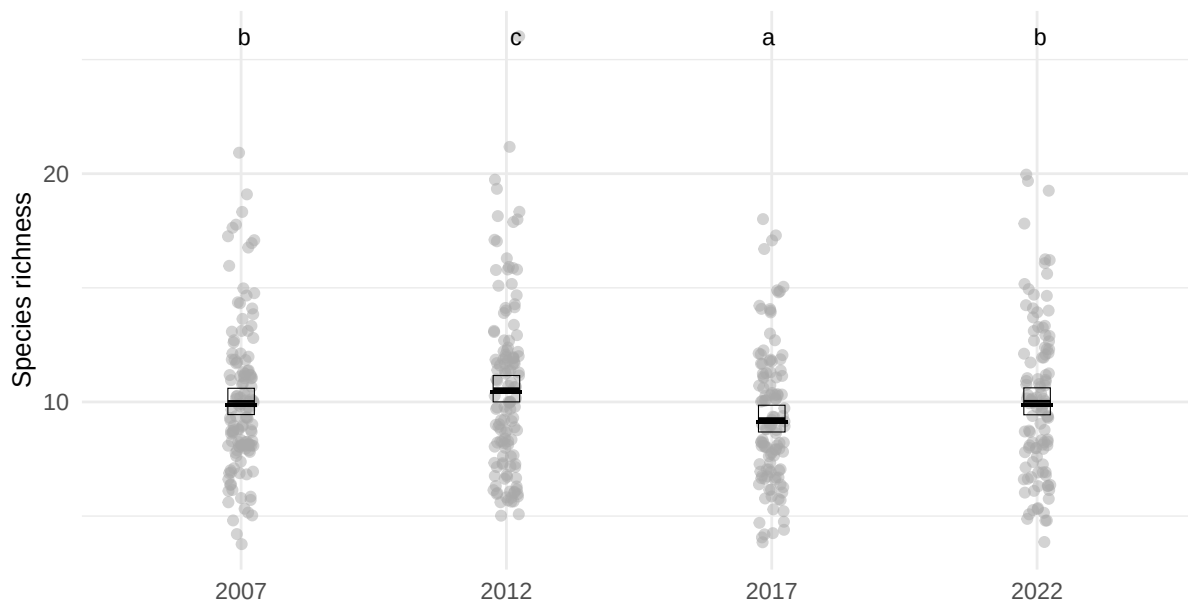


Figure 8: Species richness, number of species pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Evenness

```
model_evenness_sub_f <- lmer(
  evenness ~ factor(year) + (1 | subsection),
  data = subsection_indices
)

summary(model_evenness_sub_f)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: evenness ~ factor(year) + (1 | subsection)
Data: subsection_indices
```

```
REML criterion at convergence: -2413
```

```
Scaled residuals:
```

```
      Min       1Q   Median       3Q      Max
-3.8146 -0.5175  0.0310  0.5727  2.9727
```

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
	subsection (Intercept)	0.0001949	0.01396
	Residual	0.0002859	0.01691

Number of obs: 492, groups: subsection, 134

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	0.936409	0.001905	491.657
factor(year)2012	0.010617	0.002097	5.063
factor(year)2017	0.014378	0.002165	6.642
factor(year)2022	0.014919	0.002201	6.777

Correlation of Fixed Effects:

	(Intr)	f()2012	f()2017
fctr(y)2012	-0.544		
fctr(y)2017	-0.527	0.483	
fctr(y)2022	-0.519	0.475	0.474

```
evenness_emmeans_f <- emmeans(model_evenness_sub_f, ~ year)

evenness_pairs_f <- emmeans(model_evenness_sub_f, ~ year) |>
  pairs()

evenness_emmeans_f
```

year	emmean	SE	df	lower.CL	upper.CL
2007	0.9364	0.00190	330.2	0.9327	0.9402
2012	0.9470	0.00192	332.8	0.9433	0.9508
2017	0.9508	0.00199	355.7	0.9469	0.9547
2022	0.9513	0.00203	368.5	0.9473	0.9553

Degrees-of-freedom method: satterthwaite
Confidence level used: 0.95

```
evenness_pairs_f
```

contrast	estimate	SE	df	t.ratio	p.value
year2007 - year2012	-0.010617	0.00210	352	-5.063	<0.0001
year2007 - year2017	-0.014378	0.00216	358	-6.642	<0.0001
year2007 - year2022	-0.014919	0.00220	360	-6.777	<0.0001
year2012 - year2017	-0.003762	0.00217	355	-1.736	0.3065

```
year2012 - year2022 -0.004303 0.00220 356 -1.953 0.2080
year2017 - year2022 -0.000541 0.00224 351 -0.242 0.9950
```

Degrees-of-freedom method: satterthwaite

P value adjustment: tukey method for comparing a family of 4 estimates

```
evenness_pairs_df <- as.data.frame(evenness_pairs_f)
```

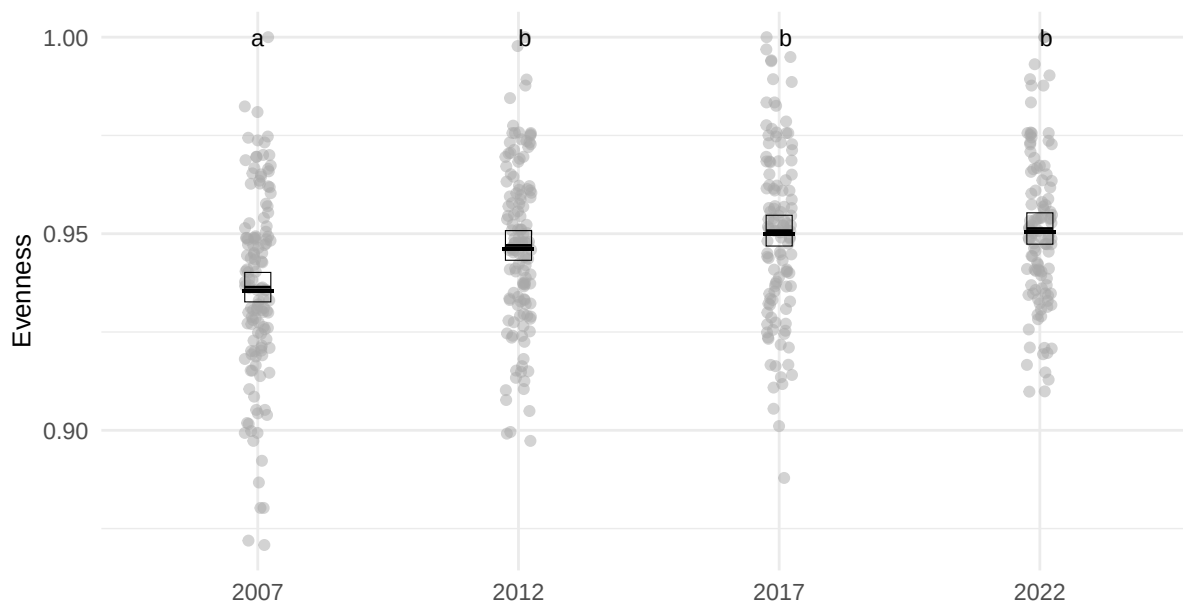


Figure 9: Evenness pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Shannon

```
model_shannon_sub_f <- lmer(
  shannon ~ factor(year) + (1 | subsection),
  data = subsection_indices
)

summary(model_shannon_sub_f)
```

Linear mixed model fit by REML ['lmerMod']

Formula: shannon ~ factor(year) + (1 | subsection)

Data: subsection_indices

REML criterion at convergence: -245.7

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.7029	-0.4843	0.0323	0.5083	3.8910

Random effects:

Groups	Name	Variance	Std.Dev.
subsection	(Intercept)	0.08371	0.2893
	Residual	0.01481	0.1217

Number of obs: 492, groups: subsection, 134

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	2.11191	0.02716	77.772
factor(year)2012	0.06457	0.01514	4.265
factor(year)2017	-0.04499	0.01569	-2.867
factor(year)2022	0.02751	0.01598	1.722

Correlation of Fixed Effects:

	(Intr)	f()2012	f()2017
fctr(y)2012	-0.274		
fctr(y)2017	-0.265	0.484	
fctr(y)2022	-0.260	0.476	0.480

```
shannon_emmeans_f <- emmeans(model_shannon_sub_f, ~ year)

shannon_pairs_f <- emmeans(model_shannon_sub_f, ~ year) |>
  pairs()

shannon_emmeans_f
```

year	emmean	SE	df	lower.CL	upper.CL
2007	2.11	0.0272	167	2.06	2.17
2012	2.18	0.0272	168	2.12	2.23
2017	2.07	0.0275	175	2.01	2.12
2022	2.14	0.0277	179	2.08	2.19

Degrees-of-freedom method: satterthwaite

Confidence level used: 0.95

`shannon_pairs_f`

contrast	estimate	SE	df	t.ratio	p.value
year2007 - year2012	-0.0646	0.0151	357	-4.265	0.0001
year2007 - year2017	0.0450	0.0157	358	2.867	0.0227
year2007 - year2022	-0.0275	0.0160	358	-1.722	0.3136
year2012 - year2017	0.1096	0.0157	357	6.994	<0.0001
year2012 - year2022	0.0371	0.0159	357	2.324	0.0945
year2017 - year2022	-0.0725	0.0161	356	-4.491	<0.0001

Degrees-of-freedom method: satterthwaite

P value adjustment: tukey method for comparing a family of 4 estimates

```
shannon_pairs_df <- as.data.frame(shannon_pairs_f)
```

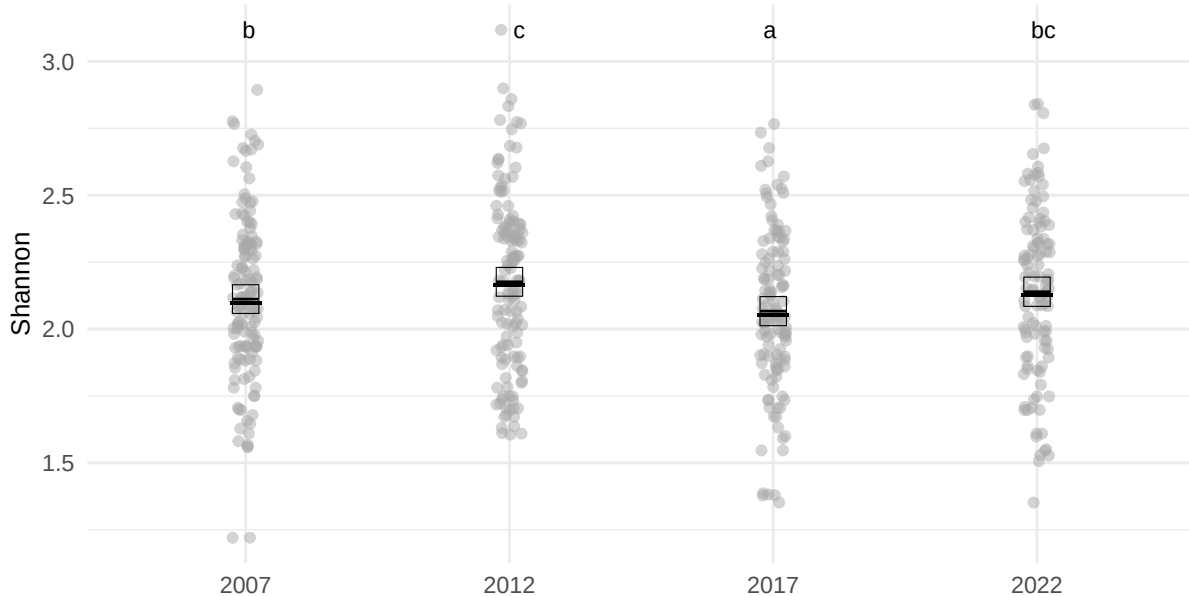


Figure 10: Shannon diversity pr. year calculate pr subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Turnover

```
# Check number of unique years per plot
plot_years <- subsection_species |>
```

```

group_by(subsection) |>
summarise(n_years = n_distinct(year), .groups = "drop")

# Filter plots with at least 2 years (turnover requires temporal comparison)
valid_plots <- plot_years |>
  filter(n_years > 1) |>
  pull(subsection)

# Filter data to valid plots only
species_long_filtered <- subsection_species |>
  filter(subsection %in% valid_plots)

# Calculate turnover per plot between years
turnover_results <- turnover(
  df = species_long_filtered,
  time.var = "year",
  species.var = "taxon_code",
  abundance.var = "foo",
  replicate.var = "subsection",
  metric = "total"
)

m_turnover_sub_f <- lmer(total ~ factor(year) + (1|subsection),
  data = turnover_results)

summary(m_turnover_sub_f)

```

Linear mixed model fit by REML ['lmerMod']
 Formula: total ~ factor(year) + (1 | subsection)
 Data: turnover_results

REML criterion at convergence: -451.7

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-2.11602	-0.58928	0.01139	0.56350	2.53936

Random effects:

Groups	Name	Variance	Std.Dev.
subsection	(Intercept)	0.008066	0.08981
	Residual	0.010461	0.10228

Number of obs: 358, groups: subsection, 131

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	0.20204	0.01199	16.844
factor(year)2017	0.03189	0.01323	2.412
factor(year)2022	-0.04771	0.01346	-3.544

Correlation of Fixed Effects:

	(Intr)	f()	201
fctr(y)2017	-0.520		
fctr(y)2022	-0.511	0.483	

```
turnover_emmeans <- emmeans(m_turnover_sub_f, ~ year) |>
  pairs()

turnover_emmeans
```

contrast	estimate	SE	df	t.ratio	p.value
year2012 - year2017	-0.0319	0.0132	238	-2.412	0.0438
year2012 - year2022	0.0477	0.0135	240	3.544	0.0014
year2017 - year2022	0.0796	0.0136	231	5.869	<0.0001

Degrees-of-freedom method: satterthwaite

P value adjustment: tukey method for comparing a family of 3 estimates

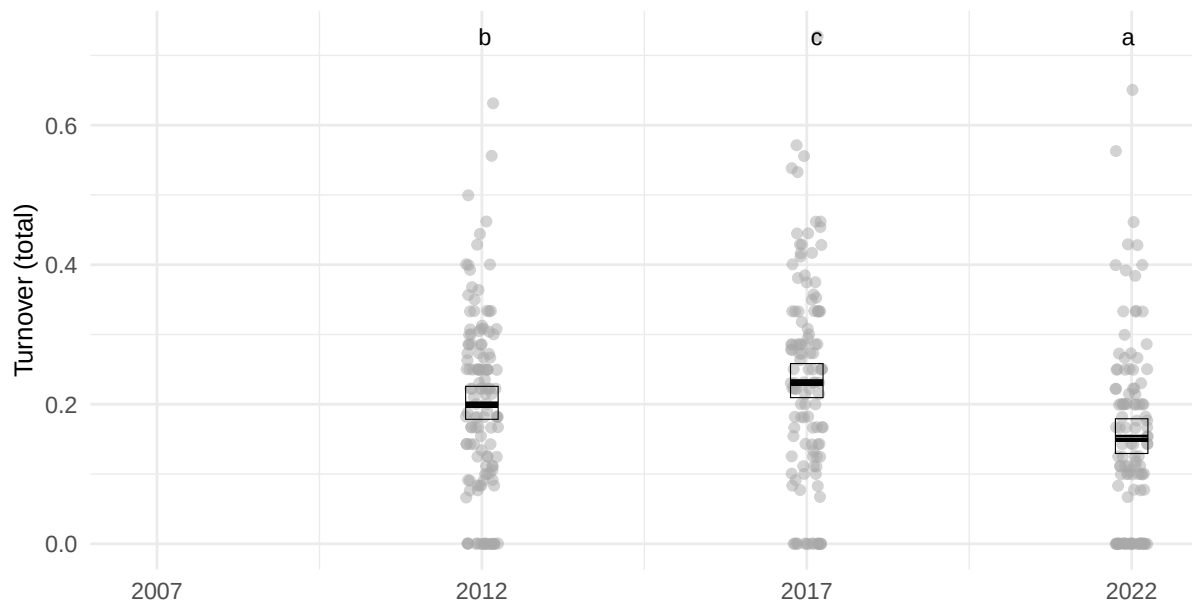


Figure 11: Turnover pr. year calculate pr subsection, but only pr present all survey years. A total of 131 subsections maintain more than 1 plots for all 4 surveys. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Plots combined

Warning: Removed 1 row containing missing values or values outside the scale range (``geom_crossbar()``).

Warning: Removed 1 row containing missing values or values outside the scale range (``geom_point()``).

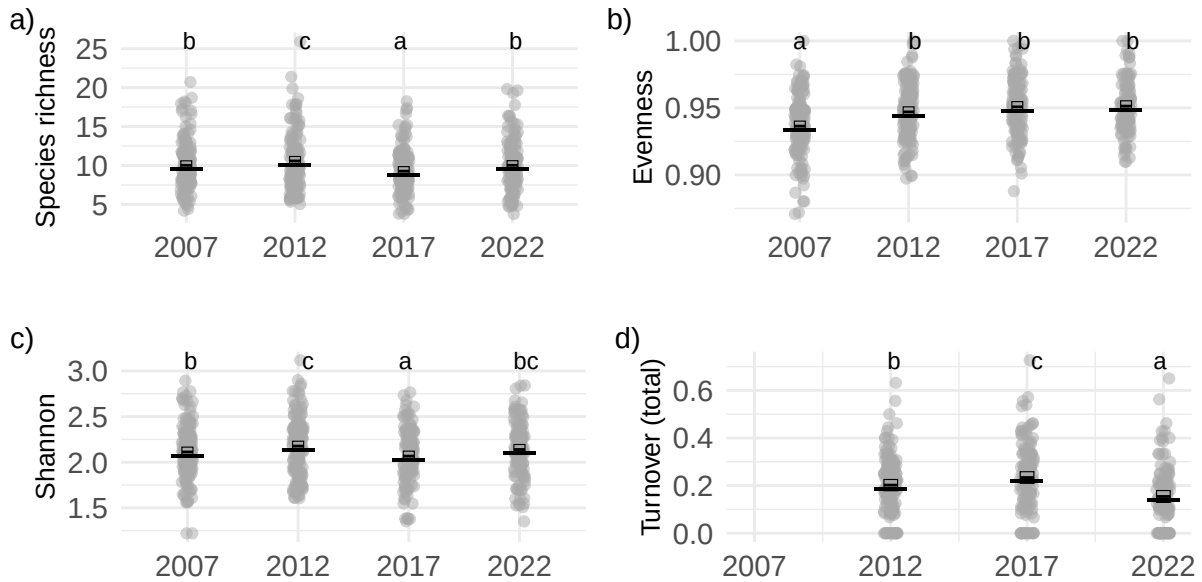


Figure 12: Diversity metrics, a) Shannon diversity, b) Species richness, c) Evenness, c) Species turnover. Shared letters indicate non-significant differences among estimated means according to compact letter display.

3-2 Community composition

```
community_matrix <- subsection_species_wide |>
  unite("sub_year_vt", subsection, year, veg_type, sep = "_", remove = FALSE) |>
  as.data.frame() |>
  column_to_rownames(var = "sub_year_vt") |>
  dplyr::select(-year, -subsection, -veg_type, -no_plots)

meta <- subsection_species_wide |>
  mutate(sub_year_vt = paste(subsection, year, veg_type, sep = "_")) |>
  dplyr::select(sub_year_vt, year, veg_type, subsection) |>
  distinct()
```

NMDS

Community composition data ready for analysis consists of 134 subsections from 4 years of surveys. 99 species have been observed across all surveys.

```

Call:
metaMDS(comm = community_matrix, k = 3, trymax = 100)

global Multidimensional Scaling using monoMDS

Data:      community_matrix
Distance: bray

Dimensions: 3
Stress:    0.1317335
Stress type 1, weak ties
Best solution was repeated 1 time in 20 tries
The best solution was from try 20 (random start)
Scaling: centring, PC rotation, halfchange scaling
Species: expanded scores based on 'community_matrix'

```

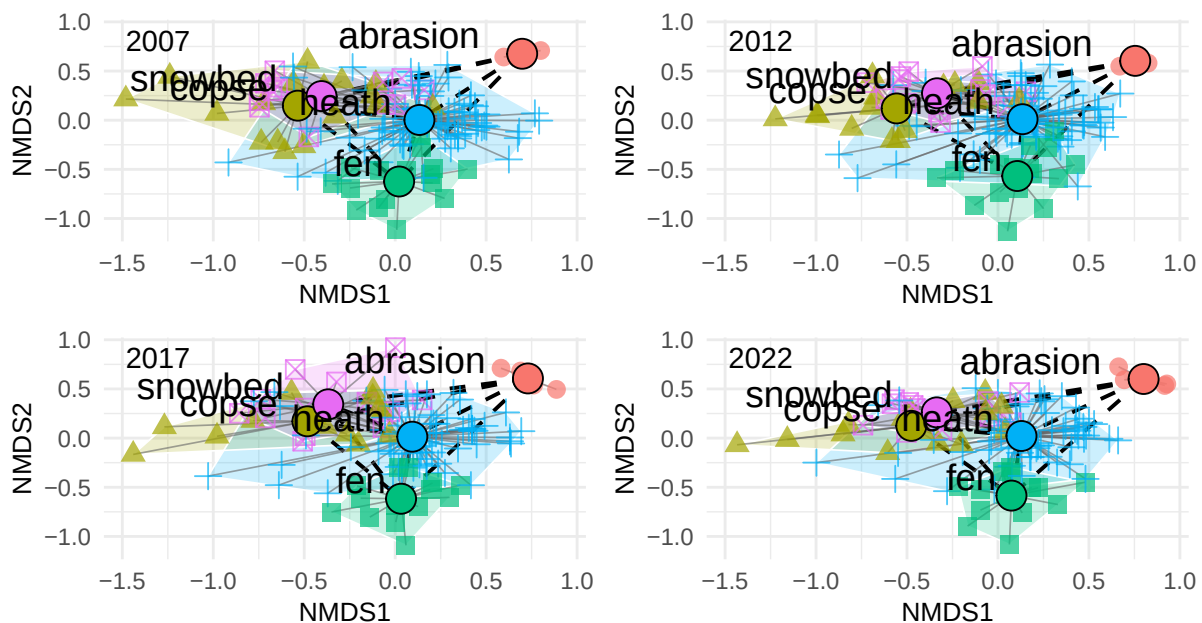


Figure 13: Plant community dynamics. Non-metric multidimensional scaling (NMDS) ordination of vegetation composition across years and vegetation types. Each panel represents subset of years of the full NMDS including all data to keep NMDS distances comparable. Grey lines connect plots within vegetation types, the within-type centroid distances. Black dashed lines indicate distances between vegetation-type centroids.

Between vegetation composition

Model: `dist ~ factor(year) + (1 | veg_pair)`, fitted by REML

N observations: 40 | N groups (veg_pair): 10

Table 1: Fixed effects: centroid distance ~ survey year (factor)

term	estimate	std.error	statistic
(Intercept)	0.886	0.119	7.462
Year 2012	-0.014	0.017	-0.780
Year 2017	-0.018	0.017	-1.024
Year 2022	-0.025	0.017	-1.448

Table 2: Estimated mean centroid distances per survey

year	emmean	SE	lower.CL	upper.CL
2007	0.886	0.119	0.618	1.154
2012	0.873	0.119	0.605	1.140
2017	0.868	0.119	0.600	1.136
2022	0.861	0.119	0.593	1.129

Table 3: Pairwise contrasts between surveys (Sidak-adjusted)

contrast	estimate	SE	t.ratio	p.value
year2007 - year2012	0.014	0.017	0.780	0.970
year2007 - year2017	0.018	0.017	1.024	0.897
year2007 - year2022	0.025	0.017	1.448	0.647
year2012 - year2017	0.004	0.017	0.244	1.000
year2012 - year2022	0.012	0.017	0.668	0.986
year2017 - year2022	0.007	0.017	0.424	0.999

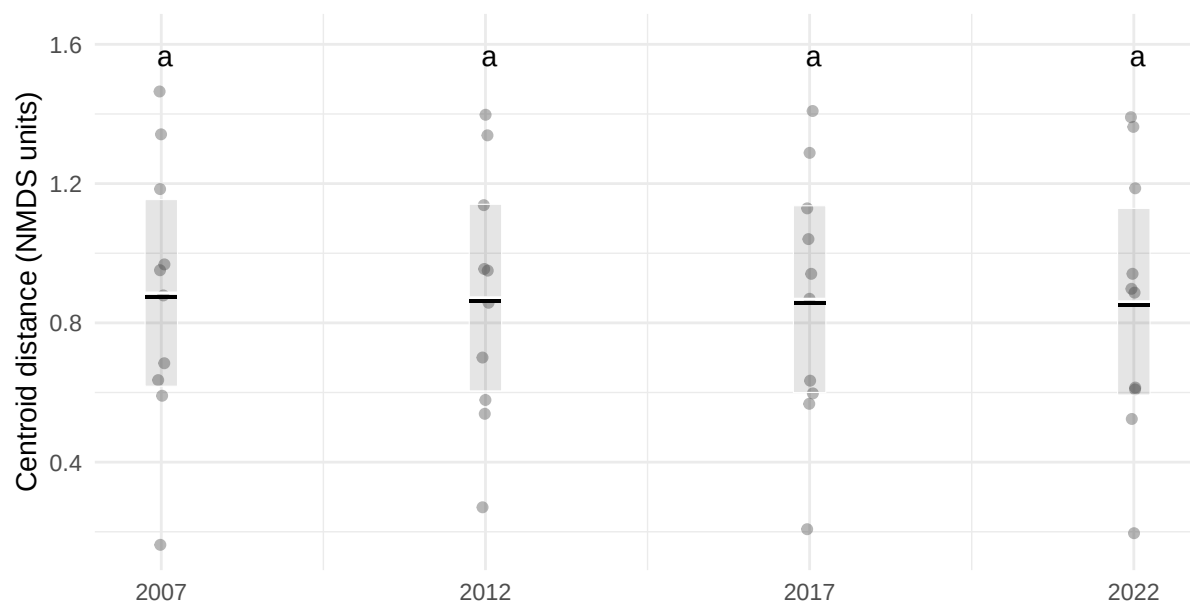


Figure 14: Between plant community dynamics. Estimated mean distance between vegetation-type centroids within each survey year based on linear mixed-effects models. Black points show estimated means ($\pm$ SE), and grey dots represent raw subsection-level distances (in NMDS units). Shared letters indicate non-significant differences among estimated means according to compact letter display.

Within vegetation composition

3-4 Functional types

Functional groups

Specific species

4 Phenology (discussion)

Data available from <https://data.g-e-m.dk> (GEM 2026b, 2026c).

NDVI

Computing bootstrap confidence intervals ...

6 warning(s): Model failed to converge with max|grad| = 0.00203777 (tol = 0.002, component 1).
See ?lme4::convergence and ?lme4::troubleshooting. (and others)

Computing bootstrap confidence intervals ...

4 warning(s): Model failed to converge with max|grad| = 0.00206721 (tol = 0.002, component 1).
See ?lme4::convergence and ?lme4::troubleshooting. (and others)

Table 4: Fixed effects

term	estimate	std.error	statistic	lower	upper
(Intercept)	0.4134	0.0348	11.8820	0.3421	0.4823
year_c	0.0010	0.0004	2.6589	0.0002	0.0018

Table 5: Random effects (variance components)

group	term	estimate
section:plot_id	sd__(Intercept)	0.0385
plot_id	sd__(Intercept)	0.1544
Residual	sd__Observation	0.0677

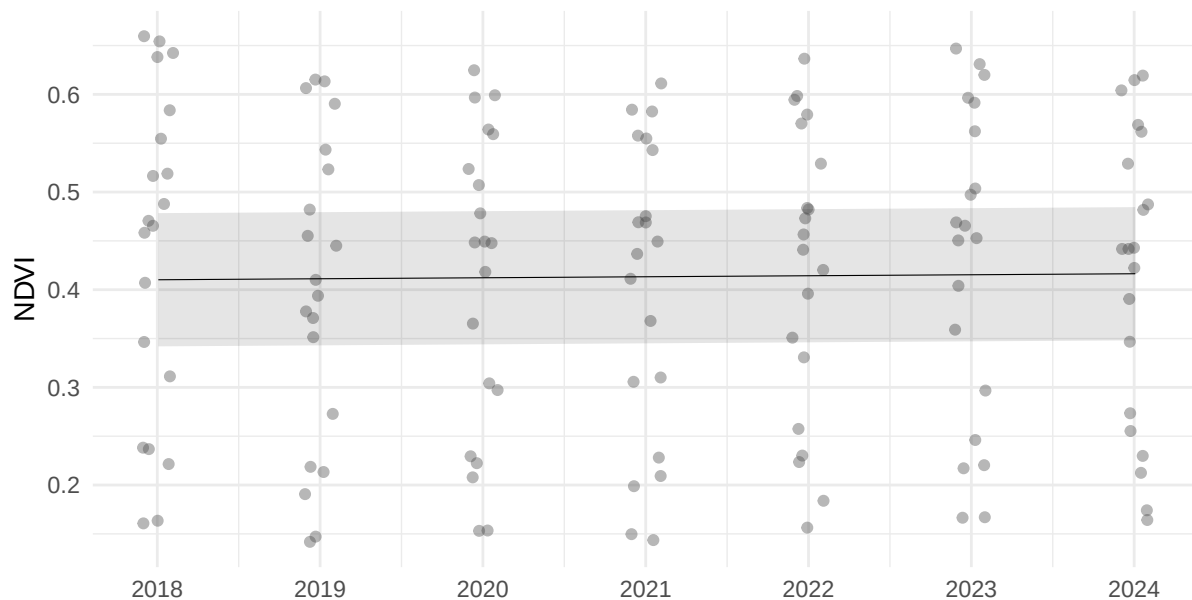


Figure 15: NDVI change over time. Mean NDVI per plot per year, based on drone measurements from 2018 to 2024. The trend line shows a significant increase of 0.001 NDVI units per year (95% CI: $3\text{e-}04$ –0.0018), representing a cumulative increase of 0.0061 NDVI units over the study period.

Salix reproductive phenology

5 References

- Cappelen, John. 2021. *Greenland – DMI Historical Climate Data Collection 1784-2020*. Nos. 21–04. Danish Meteorological Institute. <https://www.dmi.dk/fileadmin/Rapporter/2021/DMIREp21-04.pdf>.
- GEM. 2026a. “BioBasis Nuuk - Vegetation - NERO Line Raunkiaer.” Version 1.0. Greenland Ecosystem Monitoring. <https://doi.org/10.17897/HXJY-RQ13>.
- GEM. 2026b. “BioBasis Nuuk - Vegetation - Plot NDVI.” Version 1.0. Greenland Ecosystem Monitoring. <https://doi.org/10.17897/TQTT-EV69>.
- GEM. 2026c. “BioBasis Nuuk - Vegetation - Total Count (Eriophorum Angustifolium, Loiseleuria Procumbens, Salix Glauca, Silene Acaulis).” Version 1.0. Greenland Ecosystem Monitoring. <https://doi.org/10.17897/J6EK-FR47>.

GEM. 2026d. “ClimateBasis Nuuk - Precipitation - Precipitation Accumulated (Mm).” Version 1.0. Greenland Ecosystem Monitoring. <https://doi.org/10.17897/SXJ8-WA79>.

GEM. 2026e. “ClimateBasis Nuuk - Radiation - Photosyntetic Active Radiation @ 200 Cm - 5min Average (Mmol/M2/Sec).” Version 1.0. Greenland Ecosystem Monitoring. <https://doi.org/10.17897/8Z2W-D993>.

GEM. 2026f. “ClimateBasis Nuuk - Temperature - Air Temperature @ 200 Cm - 30min Average (°C).” Version 1.0. Greenland Ecosystem Monitoring. <https://doi.org/10.17897/PGN3-7597>.